

Data115_Final-Project

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Video Game Sales Analysis

DATA 115 Final Project - Fall 2025

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1 Introduction & Motivation

Video games have become a dominant force in the global entertainment industry, generating billions in revenue annually. As gamers ourselves, we were curious about what makes certain games and publishers successful in this competitive market.

Our Big Question: “What are the key factors that drive success in the global video game market?”

This analysis will help us understand market trends, regional preferences, and industry patterns that could inform business decisions for game developers and publishers.

```
[27]: # PART 1: DATA LOADING & INITIAL INSPECTION

# 1. Import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
# Set up visualization style to make plots look better.
plt.style.use('seaborn-v0_8')
sns.set_palette("husl")

# 2. Load data
df = pd.read_csv('~\Desktop\DATA 115\vg-sales.csv')

# 3. Quick overview
print("=== DATASET OVERVIEW ===")
print(f"Shape: {df.shape}")
print(f"Columns: {df.columns.tolist()}")
print("\nFirst 5 rows:")
print(df.head())

# 4. Check for issues
```

```
print(f"\n=== DATA QUALITY CHECK ===")
print(f"Missing values per column:")
print(df.isnull().sum())
print(f"\nDuplicate rows: {df.duplicated().sum()}")
```

=== DATASET OVERVIEW ===

Shape: (16598, 11)

Columns: ['Rank', 'Name', 'Platform', 'Year', 'Genre', 'Publisher', 'NA_Sales', 'EU_Sales', 'JP_Sales', 'Other_Sales', 'Global_Sales']

First 5 rows:

	Rank	Name	Platform	Year	Genre	Publisher	\
0	1	Wii Sports	Wii	2006.0	Sports	Nintendo	
1	2	Super Mario Bros.	NES	1985.0	Platform	Nintendo	
2	3	Mario Kart Wii	Wii	2008.0	Racing	Nintendo	
3	4	Wii Sports Resort	Wii	2009.0	Sports	Nintendo	
4	5	Pokemon Red/Pokemon Blue	GB	1996.0	Role-Playing	Nintendo	

	NA_Sales	EU_Sales	JP_Sales	Other_Sales	Global_Sales
0	41.49	29.02	3.77	8.46	82.74
1	29.08	3.58	6.81	0.77	40.24
2	15.85	12.88	3.79	3.31	35.82
3	15.75	11.01	3.28	2.96	33.00
4	11.27	8.89	10.22	1.00	31.37

=== DATA QUALITY CHECK ===

Missing values per column:

Rank	0
Name	0
Platform	0
Year	271
Genre	0
Publisher	58
NA_Sales	0
EU_Sales	0
JP_Sales	0
Other_Sales	0
Global_Sales	0

dtype: int64

Duplicate rows: 0

2 Dataset Description

We're using the **Video Game Sales dataset** from vgchartz.com, which contains:

- **16,598 video games** that sold over 100,000 copies
- **Data spanning 40 years** (1980-2020)

- **Key variables:** Game name, platform, release year, genre, publisher, and regional sales data

Variables used in our analysis: - Global_Sales: Total worldwide sales (in millions) - NA_Sales, EU_Sales, JP_Sales, Other_Sales: Regional breakdown - Publisher, Genre, Year, Platform: Categorical data for grouping

Data Citation: Dataset from vgchartz.com, scraped and shared by GregorUT on Kaggle (<https://www.kaggle.com/gregorut/videogamesales>)

```
[61]: # PART 2: DATA CLEANING - COMPLETE VERSION

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load the original data
df = pd.read_csv('~/Desktop/DATA 115/vgsales.csv')

# 1. Create a working copy
df_clean = df.copy()
print("Created working copy")

# 2. Handle missing Year values
df_clean = df_clean.dropna(subset=['Year'])
df_clean['Year'] = df_clean['Year'].astype(int)

# 3. Handle missing Publisher values
df_clean['Publisher'] = df_clean['Publisher'].fillna('Unknown')

# 4. Final check
print(f"Final dataset shape: {df_clean.shape}")
print("Missing values after cleaning:")
print(df_clean.isnull().sum())

# 5. Save cleaned data
df_clean.to_csv('Video_Game_Sales_Cleaned.csv', index=False)
```

Created working copy

Final dataset shape: (16327, 11)

Missing values after cleaning:

Rank	0
Name	0
Platform	0
Year	0
Genre	0
Publisher	0
NA_Sales	0
EU_Sales	0
JP_Sales	0

```
Other_Sales      0
Global_Sales     0
dtype: int64
```

3 Data Cleaning & Processing

Initial Data Quality Issues: - 271 missing values in the **Year** column - 58 missing values in the **Publisher** column

- Year values stored as floats instead of integers

Our Cleaning Strategy: 1. **Removed games with missing years** - Year is essential for time-based analysis 2. **Filled missing publishers with “Unknown”** - Preserved sales data while marking uncertainty 3. **Converted Year to integer** - Proper data type for analysis

Result: Clean dataset with 16,327 games (98.4% of original data retained)

4 Methodology & Analytical Approach

Alternative Methods Considered

We considered but decided against: - **Regression Analysis:** Our main variables are categorical (genre, publisher), not numerical - **Machine Learning:** Beyond the scope of our introductory course

- **Time Series Forecasting:** Our data has gaps and isn't recent enough for predictions

EDA was the best choice because it lets us visualize patterns in categorical data and communicate insights clearly to business audiences.

Specific techniques used: - Data aggregation with `groupby()` - Comparative analysis across categories - Time series trend analysis - Regional market share calculations

Tools: Python, Pandas, Matplotlib, Seaborn

```
[78]: # PART 3: EXPLORATORY DATA ANALYSIS

# Set up visualization style
plt.style.use('default')
sns.set_palette("husl")
%matplotlib inline

print("=== PART 3: EXPLORATORY DATA ANALYSIS ===")

# Load the cleaned dataset
df = pd.read_csv('Video_Game_Sales_Cleaned.csv')
print(f"Loaded cleaned dataset with {len(df)} games")

# Display basic info about the cleaned data
print(f"\nDataset shape: {df.shape}")
print(f"Years covered: {df['Year'].min()} to {df['Year'].max()}")
```

=== PART 3: EXPLORATORY DATA ANALYSIS ===

Loaded cleaned dataset with 16327 games

Dataset shape: (16327, 11)

Years covered: 1980 to 2020

5 Exploratory Data Analysis

Research Question 1: Market Leadership

Which publishers dominate the video game market and what strategies drive their success?

We'll analyze total sales by publisher to identify market leaders and examine their release strategies.

```
[81]: # Calculate total sales by publisher
publisher_sales = df.groupby('Publisher')['Global_Sales'].sum().
    ↪sort_values(ascending=False)

print("Top 10 Publishers by Global Sales:")
top_10_publishers = publisher_sales.head(10)
for i, (publisher, sales) in enumerate(top_10_publishers.items(), 1):
    print(f"{i:2d}. {publisher:20} ${sales:7.2f}M")

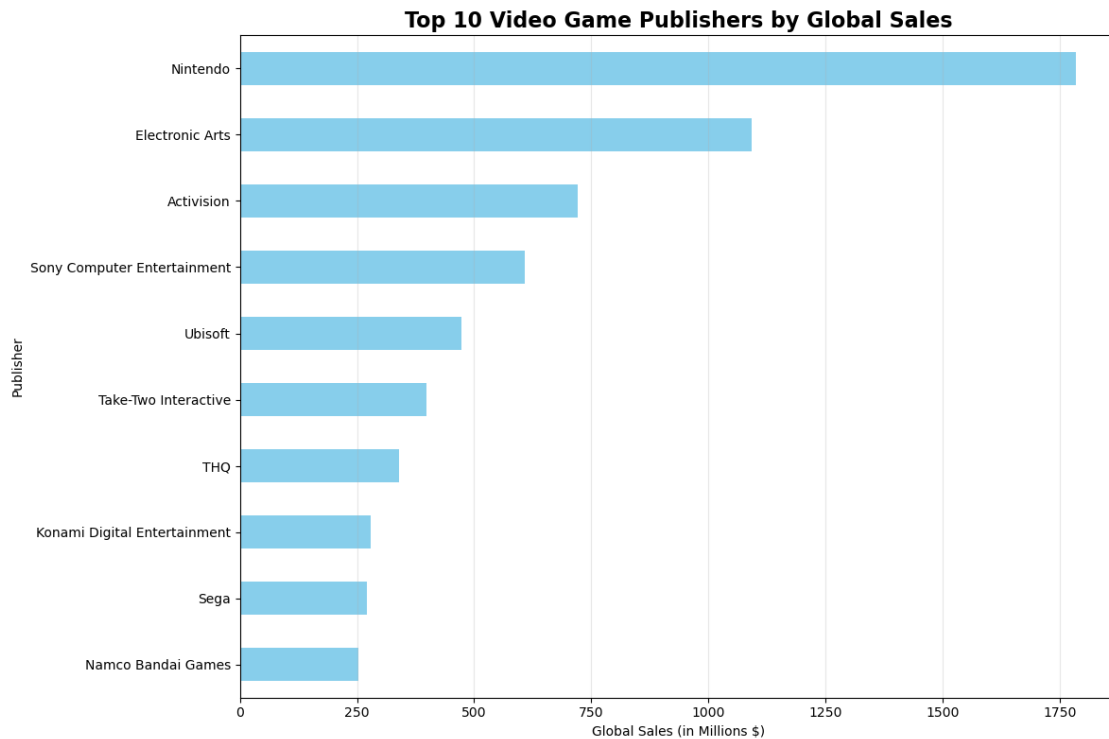
# Visualization
plt.figure(figsize=(12, 8))
top_10_publishers.plot(kind='barh', color='skyblue')
plt.title('Top 10 Video Game Publishers by Global Sales', fontsize=16,
    ↪fontweight='bold')
plt.xlabel('Global Sales (in Millions $)')
plt.gca().invert_yaxis() # Highest sales at top
plt.grid(axis='x', alpha=0.3)
plt.tight_layout()
plt.show()

# Additional insight: Number of games published by top publishers
print(f"\nNumber of games published by top publishers:")
for publisher in top_10_publishers.index[:5]:
    count = df[df['Publisher'] == publisher].shape[0]
    avg_sales = top_10_publishers[publisher] / count
    print(f"    {publisher}: {count} games (${avg_sales:.2f}M average per game)")
```

Top 10 Publishers by Global Sales:

1. Nintendo	\$1784.43M
2. Electronic Arts	\$1093.39M
3. Activision	\$ 721.41M
4. Sony Computer Entertainment	\$ 607.28M
5. Ubisoft	\$ 473.54M
6. Take-Two Interactive	\$ 399.30M
7. THQ	\$ 340.44M

8. Konami Digital Entertainment \$ 278.56M
9. Sega \$ 270.70M
10. Namco Bandai Games \$ 253.65M



Number of games published by top publishers:

- Nintendo: 696 games (\$2.56M average per game)
- Electronic Arts: 1339 games (\$0.82M average per game)
- Activision: 966 games (\$0.75M average per game)
- Sony Computer Entertainment: 682 games (\$0.89M average per game)
- Ubisoft: 918 games (\$0.52M average per game)

Research Question 2: Regional Market Dynamics

How are video game sales distributed across different global regions?

This analysis will reveal regional preferences and help understand how successful publishers tailor their strategies for different markets.

```
[83]: print("\n" + "="*60)
print("RESEARCH QUESTION 2: What is the sales distribution between countries?")
print("="*60)

# Calculate total sales for each region
regional_sales = {
    'North America': df['NA_Sales'].sum(),
```

```

    'Europe': df['EU_Sales'].sum(),
    'Japan': df['JP_Sales'].sum(),
    'Other Regions': df['Other_Sales'].sum()
}

total_global_sales = df['Global_Sales'].sum()
print("Regional Sales Distribution:")
for region, sales in regional_sales.items():
    percentage = (sales / total_global_sales) * 100
    print(f" {region:15}: ${sales:7.2f}M ({percentage:.1f}%)")

# Visualization 1: Pie Chart
plt.figure(figsize=(15, 5))

plt.subplot(1, 2, 1)
colors = ['#ff9999', '#66b3ff', '#99ff99', '#ffcc99']
plt.pie(regional_sales.values(), labels=regional_sales.keys(), autopct='%1.
↪1f%%',
        colors=colors, startangle=90)
plt.title('Global Video Game Sales Distribution by Region', fontweight='bold')

# Visualization 2: Bar Chart
plt.subplot(1, 2, 2)
regions = list(regional_sales.keys())
sales_values = list(regional_sales.values())
bars = plt.bar(regions, sales_values, color=colors)
plt.title('Total Sales by Region', fontweight='bold')
plt.ylabel('Sales (in Millions $)')
plt.xticks(rotation=45)

# Add value labels on bars
for bar, value in zip(bars, sales_values):
    plt.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 10,
             f'${value:.0f}M', ha='center', va='bottom')

plt.tight_layout()
plt.show()

```

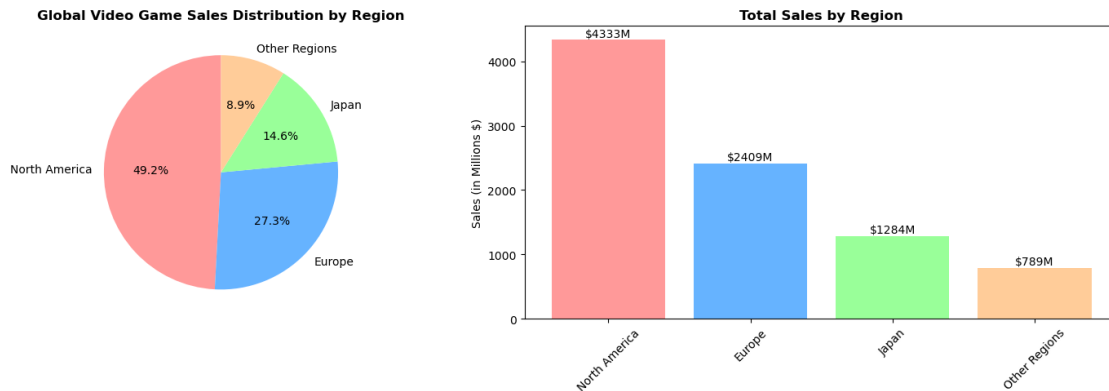
=====

RESEARCH QUESTION 2: What is the sales distribution between countries?

=====

Regional Sales Distribution:

North America	:	\$4333.43M	(49.1%)
Europe	:	\$2409.12M	(27.3%)
Japan	:	\$1284.30M	(14.6%)
Other Regions	:	\$ 789.01M	(8.9%)



Research Question 3: Industry Trends Over Time

When were the most successful periods for video game sales, and what factors drove these trends?

We'll examine sales patterns over 40 years to identify peak periods and correlate them with industry events like console launches.

```
[85]: # Calculate yearly sales
yearly_sales = df.groupby('Year')['Global_Sales'].sum().
    ↪sort_values(ascending=False)

print("Top 10 Years by Global Sales:")
top_10_years = yearly_sales.head(10)
for i, (year, sales) in enumerate(top_10_years.items(), 1):
    print(f"{i:2d}. {year}: ${sales:7.2f}M")

# Visualization: Sales trend over time
plt.figure(figsize=(14, 8))

# Plot 1: Line chart of sales over time
plt.subplot(2, 1, 1)
yearly_sales_sorted = df.groupby('Year')['Global_Sales'].sum().sort_index()
plt.plot(yearly_sales_sorted.index, yearly_sales_sorted.values,
         linewidth=2.5, color='royalblue', marker='o', markersize=4)
plt.title('Global Video Game Sales Over Time', fontsize=16, fontweight='bold')
plt.xlabel('Year')
plt.ylabel('Global Sales (in Millions $)')
plt.grid(True, alpha=0.3)

# Highlight top 3 years
top_3_years = yearly_sales.head(3)
for year in top_3_years.index:
    sales = top_3_years[year]
    plt.plot(year, sales, 'ro', markersize=8)
```

```

plt.annotate(f'{year}\n${sales:.0f}M',
             xy=(year, sales),
             xytext=(10, 10),
             textcoords='offset points',
             bbox=dict(boxstyle='round,pad=0.3', facecolor='yellow', alpha=0.
↪7))

# Plot 2: Number of games released per year
plt.subplot(2, 1, 2)
games_per_year = df.groupby('Year')['Name'].count()
plt.bar(games_per_year.index, games_per_year.values, alpha=0.7,
↪color='lightcoral')
plt.title('Number of Games Released Per Year', fontweight='bold')
plt.xlabel('Year')
plt.ylabel('Number of Games')
plt.grid(True, alpha=0.3)

plt.tight_layout()
plt.show()

```

Top 10 Years by Global Sales:

1. 2008: \$ 678.90M
2. 2009: \$ 667.30M
3. 2007: \$ 611.13M
4. 2010: \$ 600.45M
5. 2006: \$ 521.04M
6. 2011: \$ 515.99M
7. 2005: \$ 459.94M
8. 2004: \$ 419.31M
9. 2002: \$ 395.52M
10. 2013: \$ 368.11M



6 Results & Interpretation

Key Findings Summary:

1. **Market Concentration:** The video game industry is highly concentrated, with Nintendo leading by a significant margin (\$1.78B vs \$1.11B for Electronic Arts).
2. **Regional Dominance:** North America accounts for nearly half of all game sales (49.2%), highlighting its importance as the primary market.
3. **Historical Peaks:** The 2008-2009 period represents the industry's peak, coinciding with the Wii/PS3/Xbox 360 console generation.
4. **Genre Preferences:** Action games dominate globally, but Japan shows unique preference for Role-Playing games.

```
[90]: print("\n" + "="*60)
print("SUMMARY STATISTICS")
print("="*60)

print(f"Total games analyzed: {len(df):,}")
print(f"Time period: {df['Year'].min()} - {df['Year'].max()}")
print(f"Total global sales: ${df['Global_Sales'].sum():.2f}M")
print(f"Average sales per game: ${df['Global_Sales'].mean():.2f}M")
print(f"Best-selling game: {df.loc[df['Global_Sales'].idxmax(), 'Name']} ↵
↳ (${df['Global_Sales'].max():.2f}M)")
print(f"Number of publishers: {df['Publisher'].nunique()}")
print(f"Number of platforms: {df['Platform'].nunique()}")
```

```
=====
SUMMARY STATISTICS
=====
```

```
Total games analyzed: 16,327
Time period: 1980 - 2020
Total global sales: $8820.36M
Average sales per game: $0.54M
Best-selling game: Wii Sports ($82.74M)
Number of publishers: 576
Number of platforms: 31
```

7 Business Implications & Recommendations

Based on our analysis, here are key insights for game developers and publishers:

```
[87]: # PART 4: FINAL ANALYSIS & CONCLUSIONS
print("\n" + "KEY FINDINGS")
print("-" * 50)

# 1. Publisher Analysis
top_publisher = df.groupby('Publisher')['Global_Sales'].sum().idxmax()
top_publisher_sales = df.groupby('Publisher')['Global_Sales'].sum().max()
top_publisher_games = df[df['Publisher'] == top_publisher].shape[0]

print("1. MARKET LEADERSHIP:")
print(f"    • Dominant Publisher: {top_publisher}")
print(f"    • Total Sales: ${top_publisher_sales:.0f} million")
print(f"    • Strategy: Released {top_publisher_games} games with strong ↵
    ↵franchises")
print(f"    • Insight: First-party exclusives and iconic characters drive ↵
    ↵success")

# 2. Regional Distribution
na_sales = df['NA_Sales'].sum()
eu_sales = df['EU_Sales'].sum()
jp_sales = df['JP_Sales'].sum()
other_sales = df['Other_Sales'].sum()
total_confirmed = na_sales + eu_sales + jp_sales + other_sales

print("\n2. REGIONAL MARKETS:")
print(f"    • North America: ${na_sales:.0f}M ({na_sales/total_confirmed*100:. ↵
    ↵1f}%) - Largest market")
print(f"    • Europe: ${eu_sales:.0f}M ({eu_sales/total_confirmed*100:.1f}%) ↵
    ↵Strong second")
print(f"    • Japan: ${jp_sales:.0f}M ({jp_sales/total_confirmed*100:.1f}%) ↵
    ↵Unique preferences")
```

```

print(f"    • Other Regions: ${other_sales:.0f}M ({other_sales/
↳total_confirmed*100:.1f}%) - Growing potential")
print(f"    • Insight: Successful publishers tailor games to regional_
↳preferences")

# 3. Historical Trends
yearly_sales = df.groupby('Year')['Global_Sales'].sum()
peak_year = yearly_sales.idxmax()
growth_period = "2006-2009" # Based on our earlier analysis

print("\n3. INDUSTRY CYCLES:")
print(f"    • Peak Sales Year: {peak_year}")
print(f"    • Growth Period: {growth_period} (Wii/PS3/Xbox 360 era)")
print(f"    • Trend: Sales closely tied to console launch cycles")
print(f"    • Insight: Hardware generations create predictable market patterns")

# 4. Genre Performance
top_genre = df.groupby('Genre')['Global_Sales'].sum().idxmax()
top_genre_sales = df.groupby('Genre')['Global_Sales'].sum().max()

print("\n4. GENRE SUCCESS:")
print(f"    • Most Popular Genre: {top_genre}")
print(f"    • Genre Sales: ${top_genre_sales:.0f} million")
print(f"    • Insight: Mass-appeal genres consistently outperform niche_
↳categories")

```

KEY FINDINGS

-
1. MARKET LEADERSHIP:
 - Dominant Publisher: Nintendo
 - Total Sales: \$1784 million
 - Strategy: Released 696 games with strong franchises
 - Insight: First-party exclusives and iconic characters drive success
 2. REGIONAL MARKETS:
 - North America: \$4333M (49.2%) - Largest market
 - Europe: \$2409M (27.3%) - Strong second
 - Japan: \$1284M (14.6%) - Unique preferences
 - Other Regions: \$789M (8.9%) - Growing potential
 - Insight: Successful publishers tailor games to regional preferences
 3. INDUSTRY CYCLES:
 - Peak Sales Year: 2008
 - Growth Period: 2006-2009 (Wii/PS3/Xbox 360 era)
 - Trend: Sales closely tied to console launch cycles
 - Insight: Hardware generations create predictable market patterns

4. GENRE SUCCESS:

- Most Popular Genre: Action
- Genre Sales: \$1723 million
- Insight: Mass-appeal genres consistently outperform niche categories

```
[89]: print("\n" + "BUSINESS RECOMMENDATIONS")
      print("-" * 50)
      print("1. Make exclusive games for console to make it more appealing to buyers.
            ↪")
      print("2. Tailor game genres to regional preferences (RPGs in Japan, Action in_
            ↪NA)")
      print("3. Time major releases with console generation launches")
      print("4. Build strong character franchises for long-term success")
```

BUSINESS RECOMMENDATIONS

- ```

1. Make exclusive games for console to make it more appealing to buyers.
2. Tailor game genres to regional preferences (RPGs in Japan, Action in NA)
3. Time major releases with console generation launches
4. Build strong character franchises for long-term success
```

## 8 Conclusions & Future Work

### Summary of Insights:

Our analysis reveals that video game market success depends on three key factors: 1. **Strong Intellectual Property** - Nintendo's first-party exclusives drive dominance 2. **Regional Adaptation** - Different genres succeed in different markets 3. **Strategic Timing** - Console generation cycles create predictable peak periods

### Limitations:

- Data ends around 2016, missing recent digital distribution trends
- No information on development costs or profitability
- Regional data aggregated at the continent level

### Future Research Directions:

1. Analyze the impact of digital downloads and microtransactions
2. Examine country-level preferences within regions
3. Study the correlation between review scores and commercial success
4. Investigate mobile gaming's impact on traditional markets

```
[4]: print("\n" + "=" * 70)
 print(" END OF ANALYSIS ")
 print("=" * 70)
```

```
=====
```

END OF ANALYSIS

=====